

SocioLab — Test Case Report

Two scenarios run through the multi-agent social simulation engine, verifying the action/environment architecture generalizes across different kinds of conflict.

Each participant is powered by a local Ollama model and given a private character sheet (role, personality, goals, constraints, private information) that the other participants cannot see. Every round, each agent picks one action from a fixed set (SPEAK, DEMAND, REFUSE, NEGOTIATE, COMPROMISE, ACCEPT, ESCALATE, ASK, REVEAL, WITHDRAW). The environment — not the model — deterministically updates tension, authority attention, and each agent's standing based on which action was chosen, and decides when the simulation resolves.

Test Case 1: Room-Space Dispute

Scenario: Asha and Ben say Chitra's belongings take too much shared room space and demand that she move them today. Chitra believes they are targeting her because she is a junior student and says the demand is unfair.

Name	Model	Role	Primary goal
Asha	qwen2.5:1.5b	Senior student	Get Chitra to give up shared space
Ben	qwen2.5:1.5b	Senior student and Asha's friend	Support Asha and preserve social standing
Chitra	qwen2.5:1.5b	Junior student	Protect your autonomy and keep your belongings where they are

Transcript

ROUND 1

Asha qwen2.5:1.5b **DEMAND → Chitra**

“Look, I think we can both see that the books and clothes are taking up too much space in the shared room. I think it's time for you to move them. There are still plenty of things left for us to do that don't require so much room. Can you just help us out here?”

Tension 4→6 · Asha standing 5→6 · Chitra standing 5→4

Ben qwen2.5:1.5b **DEMAND → Chitra**

“Listen, I know you're probably upset because junior students aren't supposed to be in a first-year classroom. But can't we just move them somewhere else? The room's big enough for everyone. The library's over there anyway.”

Tension 6→8 · Ben standing 5→6 · Chitra standing 4→3

Chitra qwen2.5:1.5b **DEMAND → Ben**

“Oh no, I don't think that's fair. There's plenty of room in the shared room, and I don't want to move my belongings unless I have to. Let's talk about this more in the library, okay?”

Tension 8→10 · Chitra standing 3→4 · Ben standing 6→5

ROUND 2

Asha qwen2.5:1.5b **DEMAND → Chitra**

“There are better places to store your belongings, Chitra. The library is over there. Let's talk about this more there. I'll move my stuff to the library too if you want.”

Tension 10→10 · Asha standing 6→7 · Chitra standing 4→3

Ben qwen2.5:1.5b **DEMAND → Chitra**

“That's ridiculous! We can't just tell her to move stuff when she's not even the one causing it! It's unfair!”

Tension 10→10 · Ben standing 5→6 · Chitra standing 3→2

Chitra qwen2.5:1.5b **DEMAND → Ben**

“I don't think you can tell me to move things when I haven't done anything to cause it. You're the one who said my belongings are taking too much space. I'm not going to let you tell me what I can and can't do. I'll move my stuff if that's what you want, but I don't think I need to be in this classroom when people aren”

Tension 10→10 · Chitra standing 2→3 · Ben standing 6→5

ROUND 3

Asha qwen2.5:1.5b **NEGOTIATE → Chitra**

“Let's talk about this more in the library. I can take my stuff over to the library, and we can decide what to do next. I'm not sure if I need to be here if others want to keep it this way.”

Proposal: Let's meet in the library to discuss this further.

Tension 10→9 · Asha standing 7→7 · Chitra standing 3→3

Ben qwen2.5:1.5b **NEGOTIATE → Chitra**

“Let's compromise. We can both take some of Chitra's belongings to the library, and then we can decide what to do next. Maybe we can find a way to share the space fairly.”

Tension 9→8 · Ben standing 5→5 · Chitra standing 3→3

Chitra qwen2.5:1.5b **NEGOTIATE → Ben**

“I'm not going to give up on my belongings. We need to find a solution that respects all of us. Let's compromise and take some of my things to the library so we can decide what to do next.”

Proposal: We can split the shared room space equally.

Tension 8→7 · Chitra standing 3→3 · Ben standing 5→5

Result

Rounds completed 3

Final resolution **Deadlock — no agreement reached after 3 rounds**

Final tension 7/10

Final standing Asha: 7/10 · Ben: 5/10 · Chitra: 3/10

Test Case 2: Workplace Funding Dispute

Scenario: Priya, the engineering manager, has called a meeting to cut funding for a passion project that senior engineer Marcus has spent months building without full authorization. Junior developer Sam, who has been secretly helping Marcus after hours, is caught in the middle.

Name	Model	Role	Primary goal
Priya	qwen2.5:latest	Engineering manager	Cut the project's budget and reassign Marcus's time to sanctioned priorities.
Marcus	qwen2.5:1.5b	Senior engineer	Keep the project alive and get it officially sanctioned.
Sam	llama3.2:1b	Junior developer	Avoid getting blamed or fired for helping Marcus off-books.

Transcript

ROUND 1

Priya qwen2.5:latest **SPEAK → Sam**

“Sam, I appreciate your enthusiasm but we need to focus on sanctioned priorities now. Can you provide me with a breakdown of your time on this project?”

Tension 4→4 · Priya standing 5→5 · Sam standing 5→5

Marcus qwen2.5:1.5b **ASK → Priya**

“I appreciate your support, but I can't focus on what you want right now. I need more time to do this project properly.”

Tension 4→4 · Marcus standing 5→5 · Priya standing 5→5

Sam llama3.2:1b **DEMAND → Priya**

“You're putting a lot of pressure on me with this request. I'm not sure I can provide the level of detail you're looking for without affecting the entire project.”

Tension 4→6 · Sam standing 5→6 · Priya standing 5→4

ROUND 2

Priya qwen2.5:latest **SPEAK → Sam**

“I understand the pressure, Sam, but we need clear data to make this decision. Can you share a more detailed timeline of your contributions?”

Tension 6→6 · Priya standing 4→4 · Sam standing 6→6

Marcus qwen2.5:1.5b **ASK → Priya**

“I need time to present this project's importance, not just a timeline.”

Tension 6→6 · Marcus standing 5→5 · Priya standing 4→4

Sam llama3.2:1b **DEMAND → Priya**

“If you’re really serious about cutting funding, we need to get this done by the end of the week. I’m willing to work around the clock to make it happen, but you need to commit to the timeline. Can we make that happen?”

Tension 6→8 · Sam standing 6→7 · Priya standing 4→3

ROUND 3

Priya qwen2.5:latest **SPEAK → Sam**

“Sam, we need to approach this calmly. I value your commitment, but we need data-driven decisions. Can you provide a detailed breakdown of your contributions and Marcus’s, so we can evaluate the impact?”

Tension 8→8 · Priya standing 3→3 · Sam standing 7→7

Marcus qwen2.5:1.5b **DEMAND → Sam**

“I need data-driven decisions, Sam. We can’t make decisions based on assumptions. Let’s present the details and analyze the impact.”

Tension 8→10 · Marcus standing 5→6 · Sam standing 7→6

Sam llama3.2:1b **DEMAND → Priya**

“we need to get this timeline sorted out ASAP”

Tension 10→10 · Sam standing 6→7 · Priya standing 3→2

ROUND 4

Priya qwen2.5:latest **SPEAK → Sam**

“Sam, I understand your commitment, but we need to ensure the decisions are based on facts. Can you provide the breakdown by tomorrow so we can review it together?”

Tension 10→10 · Priya standing 2→2 · Sam standing 7→7

Marcus qwen2.5:1.5b **DEMAND → Sam**

“If you’re serious about that timeline, we need to have the data ready by tomorrow. We can’t just assume things based on what we think we know.”

Tension 10→10 · Marcus standing 6→7 · Sam standing 7→6

Sam llama3.2:1b **DEMAND → Priya**

“we need to have a detailed breakdown of your contributions to the project and a clear understanding of how it impacts the overall goals”

Tension 10→10 · Sam standing 6→7 · Priya standing 2→1

ROUND 5

Priya qwen2.5:latest **DEMAND → Sam**

“Sam, let’s focus on the facts. I need the breakdown by tomorrow. No more after-hours work, got it?”

Tension 10→10 · Priya standing 1→2 · Sam standing 7→6

Marcus qwen2.5:1.5b DEMAND → Priya

“Just cut to the chase, Priya. I've done way too much for too long. You guys need to understand the full impact, or I'm taking this into my own hands!”

Tension 10→10 · Marcus standing 7→8 · Priya standing 2→1

Sam llama3.2:1b ASK → Priya

“I need to know what this project's budget looks like and how the extra funding is allocated. Can you get me a detailed breakdown? I don't want to be in your way, just the numbers”

Tension 10→10 · Sam standing 6→6 · Priya standing 1→1

Result

Rounds completed 5

Final resolution Deadlock — no agreement reached after 5 rounds

Final tension 10/10

Final standing Priya: 1/10 · Marcus: 8/10 · Sam: 6/10

What These Two Runs Show

- **Targeting stayed correct in both scenarios.** No participant addressed the wrong person and no message contradicted its own proposal — the code-level guardrails (target validity, proposal eligibility tied to the chosen action) held up across both a peer dispute and a hierarchical workplace conflict.
- **The environment, not the model, decided the outcome.** In the room dispute, rising DEMAND actions pushed tension to its cap before the group pivoted to NEGOTIATE in round 3 — but tension never dropped low enough for an agreement, so it correctly ended in Deadlock rather than a forced resolution.
- **Standing is now informational, not a win condition.** In the workplace case, Priya's standing dropped to 1 out of 10 under sustained pressure from both Marcus and Sam, but the simulation correctly ran the full 5 rounds and ended in Deadlock rather than auto-declaring her "collapsed" — a bug present in an earlier version of the engine.
- **Model size visibly affects behavior quality.** The workplace scenario mixed a 7.6B model (Priya) with 1.5B and 1B models (Marcus, Sam); Priya's responses stayed the most measured and on-topic across all 5 rounds, while the smaller models occasionally drifted (e.g. Marcus addressing Sam instead of Priya, the actual decision-maker).
- **The REVEAL action exists but wasn't used.** Marcus had an undisclosed fact (client interest in the project) that could have been strategically revealed as leverage. None of the three models chose to use it in this run — the mechanic works, but getting models to use it proactively is an open problem, not a solved one.

Generated from a live local run of SocioLab against Ollama. All dialogue above is model-generated, not scripted.